

Laws that stipulate drone / UAS

- Cap 448C (HK)
 - shall not recklessly or negligently cause or permit drone to endanger person or property
 - offenders → prosecution, fine, 2 years imprisonment
- new SUA law: Cap 448G, 1/6/2022
 - safeguard public aviation safety
 - allow room for development and innovation of SUA
 - SUA definition: < 25 kg, including everything attached
- Cannot fly in:
 - restricted flying zone (RFZ)
 - prohibition area
 - person / vehicle / structures not under control
 - areas that present a risk
- except in enclosed area

Choice of flying sites

- direct line of sight
- flat area for take-off and landing
- away from people / vehicle / structure / helipad / paragliding areas
- away from power sources (powerlines, transmitter towers)

Time of Operation

- in daylight hours only

Altitude

- A1: ≤ 100ft
- A2, B, C: ≤ 300ft

Categories

- A1: ≤ 250g
- A2: ≤ 7 kg
- B: ≤ 25 kg
- C: ≤ 150 kg

Registration

A1 A2 B

✓ ✓

Labelling

✓ ✓

Equipment

✓ ✓ (flight log & geo awareness)

Training

✓

Insurance

✓

CAD Permission

✓

Operation Requirements

	A1	A2
minimum lateral separation	10m	30m
maximum speed	20km/h	50km/h
dimensions	1m,	
		motor blade tips 1.2m

CAD Issued labels

- SA A T W W label
- standardized size and affix mechanism
- adhesive label
- tamper proof
- weather proof
- water proof

Duties of Pilots

- chief pilot and second pilot
- Chief Pilot:
 - ① overall direction and performance
 - ② command and control over SUA
 - ③ keep flight incident & maintenance records
 - ④ coordinate with client / relevant parties over safety arrangements

Second Pilot:

- ① ensure good communication to operator any turning instructions
 - ② avoid crashing obstacles
 - ③ monitoring the UAS flight parameters
 - height, distance, battery level, satellite required
- activate return home button
 - terminate operation
 - take over role of the chief pilot in case of he is incapacitated

GSD

- ground sampling distance
- distance between two pixels $\rightarrow$ ground

h : height

Imw : image width

sw : sensor width

f : focal length

$$\frac{h}{f} = \frac{GSD}{\frac{sw}{Imw}}$$

$$GSD = \frac{h \cdot sw}{f \cdot Imw}$$

GSD_h : use Imh , Sh

Georeferencing:

- RTK: data link required
- PPK (post processing kinematic)
- position accuracy 3~5 cm

Non georeferenced images

- measure the dimension of certain points
- apply the scale for 3D model generated

Procedure for UAV survey

- ① site reconnaissance
 - identify survey area for control point survey
- ② establish control points
 - use GPS / UAV with GPS

Camera model and setup

① Hardware

- any standard camera
- depends on quality of sensor and optics

② Recommendations

- constant focal length
- adjust lighting conditions and camera settings
 - avoid blurry photos
- forbid digital zoom / geometric retouches
- image format: tiff, high quality JPG

Data capture requirements

- each point at least 3 photos
- 60% overlap
- 15% max angle
- consistent distance from objects

To capture more details:

- zoom + add transition photos
- circle around at different levels
- more photos

Camera factors

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

u : object distance
 v : image distance
 f : focal length

$$v \approx f, u \rightarrow \infty, c \rightarrow f$$

- equivalent $f = \text{actual } f \cdot \text{crop factor}$

Systematic errors, factors affecting photogrammetry

- shooting distance
- baselines
- angle of incidence
- % overlap
- # overlap
- intersection angle